Modeling, Estimation, and Control of Human Circulatory System With a Left Ventricular Assist Device

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***Abstract*—In this paper, a state–space model is developed through theoretical analysis and numerical solutions to approximate the response of the human circulatory system. This system model has one critical time-varying parameter: the resistance of peripheral blood vessels. A parameter estimation scheme is derived to estimate this parameter, and the parameter estimate is used to implement an adaptive observer to estimate the aortic pressure for physiological control. An optimal adaptive controller is proposed to control the estimated aortic pressure to track a reference signal updated by a nonlinear function of the pump head to meet the physiological need. A Matlab simulation program and an experimental mock human circulatory loop are employed as test environments for the human circulatory systems with a left ventricular assist device and their physiological controllers. Different physiological conditions, such as the variation of left ventricular failures, variation of activities, and collapse of the left ventricle, are evaluated to test the designed physiological control system. Simulation and experimental results consistently show that the aortic pressure estimation error is small, and that the abnormal hemodynamic variables of a congestive heart failure patient are restored back to the normal physiological range.**

***Index Terms*—Adaptive control, circulatory system, estimation, feedback control, left ventricular assist device, modeling, pump head.**

I. Introduction

HEART disease is the leading cause of death in the United States. The traditional solution to end-stage congestive heart failure (CHF) is heart transplantation. However, some patients are not eligible for a transplant because of age or health reasons. Even if the patients are eligible for a transplant, the severely limited supply of donor hearts can give only approximately 10% of patients a transplantation each year [1]. Therefore, mechanical circulatory assist devices, called artificial heart pumps (AHPs), have been introduced to save some lives of end-stage CHF patients since the 1960s. AHPs have several advantages over heart transplants: they are less costly and not limited by the availability of heart donors, they may provide treatment for patients uneligible for heart transplants, and they may lead to the recovery of the failed ventricle.

An AHP can either be used as a total artificial heart (TAH) when it replaces both ventricles, or as a left ventricular assist device (LVAD) when it aids the failed left ventricle by supplying additional blood flow but does not replace the left ventricle. Since the left ventricle takes most of the working load for blood circulation, about 75% of heart failures are caused by predominant left ventricular failure. Therefore, LVADs are in increasing demand for heart failure patients, and their development has become an important research topic of major interest. The configuration of a LVAD in a human circulation system is the following: the inlet is connected to the left ventricle, while the outlet is connected to the aorta.

There have been three generations of AHPs: pulsatile pumps since the 1960s, rotary pumps with contact bearing since the 1980s, and rotary pumps with noncontact bearing which are currently under development. In the latest type pumps, the impeller is levitated by either magnetic or hydrodynamic forces without any mechanical contact. There is no bearing wear or bearing seal so that there is no thrombosis factor. Some studies have showed this type of pumps demonstrating an excellent hemolytic performance over some rotary pump with contact bearing [21]. Moreover, this type of pumps are expected to have a long mechanical durability to work for 10 to 20 years. The researchers at Pennsylvania State University Erie, University of Virginia, and Utah Artificial Heart Institute, among other groups in the world, are developing a new magnetically levitated axial flow heart pump (LEV VAD) as a permanent LVAD.

Currently, LVADs have been used successfully for short term (less than 2 years) as a bridge to heart transplant [2] with a constant pump speed. The long-term use of LVADs is expected to benefit end-stage CHF patients to a greater level, as a destination therapy or bridge to recovery device. Scientists and researchers have been actively studying these issues to extend the life of AHPs, such as optimizing pump design using computational fluid dynamics to increase efficiency and reduce hemolysis [22], [23], suspension of LVAD impeller with magnetic forces [21], [24], [25], effect of LVAD on natural heart function and remodeling [26], and physiological control system that will be discussed in this paper [4]–[13], [15]–[20], [27].

During the long-term support of LVAD, patients may sleep, rest, or exercise slightly, like walking or climbing stairs. Besides these activities, the recipient's left ventricle may get better or

Manuscript received August 16, 2005; revised February 6, 2006. Manuscript received in final form December 7, 2006. Recommended by Associate Editor F. Ghorbel. This work was supported by Utah Artificial Heart Institute, Department of Health and Human Services, National Institutes of Health, and the National Heart, Lung, and Blood Institute under Grant R01 HL64378-01.

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Digital Object Identifier 10.1109/TCST.2006.890288

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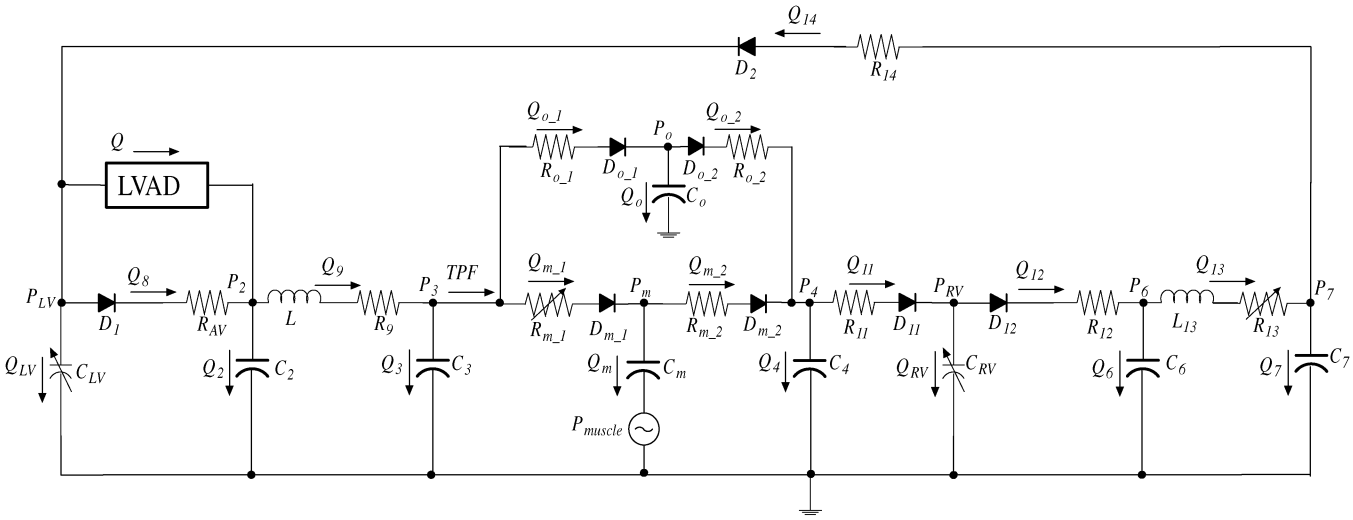


Fig. 1. Circuit analog of circulatory systems with an LVAD [4].

worse. In these conditions, the parameters of the human circulatory system may change dramatically, so does the body's need for blood. To match the body's need, the pump speed of LVADs needs to be adjusted correspondingly. Therefore, one key issue to be addressed in order to achieve this long-term use of LVADs is the development of an effective and efficient (robust and adaptive) physiological control system.

In this paper, we give an overview of the recent research on the development of physiological controllers for LVADs, and present some of our new analytical, simulation and experimental results on the modeling and control of LVAD systems. The human circulatory system was represented by a state–space model, whose only time-varying parameter is the total peripheral resistance (TPR). The value of TPR was tracked by an adaptive estimator and used to update the gains for an optimal controller. Controller performance in pathological and activity variations and left ventricular collapse were studied. The physiological controller is shown to be able to handle large TPR variations, to reject major disturbance caused by the heartbeat, and to satisfy the varying body needs in different physiological conditions. This paper forms the fundamental structure for a permanent LVAD's physiological controller, and studies the most common physiological loadings situations in longterm use.

II. Human Circulatory System Model

A lumped parameter model of human circulatory systems can be analogously represented by an 11th-order electric circuit shown in [4, Fig. 1]. It consists of analog blocks for left ventricle (subscripts LV), aorta (2), systemic artery (3), systemic vein and right atrium (4), right ventricle (RV), pulmonary artery (6), pulmonary vein and left atrium (7). The annotations and values for each element are listed in the Appendix. Two parallel blocks for skeletal muscle (m) and nonmuscular peripheral organs (o), respectively, are placed between the blocks of systemic artery and systemic vein. Resistors represent the viscous property of blood flow, while inductances embody the inertia property of blood flow. Capacitors model the elastic property, or compliance of the vessel wall, and diodes are used to mimic the properties of one-directional valve. Muscular vascular resistance (R_{m_1}) and pulmonary vascular resistance (R_{13}) are modeled as variable resistors to accommodate their variations in different activity levels. LVAD is placed between the left ventricle and the aorta, exactly where it will be placed in implantation surgery.

In this paper, the pressure and volume relation of ventricle is modeled as [4]

$$P(t) = P_s(t) = E_{\max} E_n(t)\,[V(t) - V_0] + a e^{b[V(t)-V_0]} + R_V \dot{V} \quad \text{in systole} \tag{2.1}$$

$$P(t) = P_d(t) = a e^{b[V(t)-V_0]} \quad \text{in diastole} \tag{2.2}$$

where R_V is the parameter to model the viscous property of heart muscle during systole; V_0 is the unstressed volume at which no measurable pressure is built up, in reference to extra-ventricular pressure; $E_{\max}$ is a parameter representing the contractility of ventricle; a and b are two small coefficients to model ventricular diastolic function; and $E_n(t)$ is the normalized systole elastance function

$$E_n(t) = \left[1 - (2t/t_s - 1)^2\right] \tag{2.3}$$

where t_s is the period of systole and t is the time of interest. In Fig. 1, the pressure and volume relation of the ventricle is represented by nonlinear capacitors C_{LV} and C_{RV}, where capacitance $C(t)$ defined as $[V(t) - V_0]/P(t)$.

1) Left-Sided CHF: CHF is a chronic condition in which the heart cannot pump blood at a rate which effectively supplies the body's tissues and organs [3]. Most heart failures are caused predominantly by depression of the left ventricular systolic function, and usually called left-sided CHF. When left-sided CHF becomes severe and life threatening, these patients with CHF are prime candidates for LVADs. For these patients, their right heart and the peripheral circulatory system are also different from those of a healthy person, because of the interaction between left and right heart, along with the autoregulation mechanism of the human body [4].

In this paper, left-sided CHF is modeled by the variations of the parameters for the left ventricle, the right ventricle, and the peripheral circulation. For the left ventricle's parameters in (2.1), (2.2), $E_{\max}$ was decreased, V_0 and b were increased. For right ventricle, $E_{\max}$ was decreased while b were increased, both by a smaller ratio than those of the left ventricle, to model the dependence of the right ventricular function on the left ventricular function. The heart rate was set to a higher value consistent with compensatory tachycardia. TPR were increased to model the vasorestriction effect of the human body in the presence of the blood flow drop [4].

The symptoms of LVAD candidates were successfully simulated by this model, such as: decreased ejection fraction (EF), decreased cardiac output (CO), i.e., total flow through peripheral vessels (TPF), and elevated left ventricular end-diastolic pressure (LVPED).

2) Muscle Pump in Exercise: In exercise, the skeletal muscles contract and relax alternately, thus generating a periodic intramusclar pressure, which is positive in contraction and negative in relaxation. This intramuscular pressure along with the venous valve, helps to push the blood in the veins of the working muscle back to the heart, like a booster pump. With the muscle pump function represented by the periodic pressure resource P_{muscle} (sinusoidal-like waveform) [4] and the diodes D_{m_1} and D_{m_2}, the sudden increase of central venous pressure at exercise onset was successfully simulated, in agreement with experimental results reported in [29] and [30]. Some simulation results for a healthy person showed that the cardiac output without the muscle pump underestimates the cardiac output with the muscle pump by 10% [4]. This discrepancy can be explained from the perspectives of cardiac function and peripheral circulation. On the one hand, the increase in central venous filling from the muscle pump function augments the right ventricular end-diastolic volume, thus increasing the stroke volume. On the other hand, the muscle pump function helps to decrease the effective resistance of the blood vessel in the working muscle, resulting in the increase of the peripheral blood flow, equal to the cardiac output in average. The muscle pump function is essential to simulate the circulatory response in exercise properly, which provides a better testing environment for long-term LVAD physiological control system.

III. Objectives of Physiological Control

As a destination therapy device, modulation of the degree of ventricular assist and overall cardiac output based on physiological needs are imperative for long-term LVAD performance. Control in the physiological sense of rotary LVADs presents major challenges because they create suction when the impeller speed, and thus pump flow, exceeds the venous return to the left ventricle. The level of blood flow assistance can become inadequate when the speed is too low, for example, there may be a retrograde flow of blood through the rotating impeller, from the pressurized aorta into the left ventricle, during ventricular diastole [5]. Avoiding retrograde flow when the speed is too low, and suction when the speed is too high, establishes the upper and lower limits of the LVAD rotational speed. These limits will frequently change in the patient, and particularly when the patient's heart moves more into and out of failure or when the patient starts to exercise.

To reverse the symptoms of CHF as explained before, and to avoid damage to the cardiovascular system, the main objectives for a physiological control system, as summarized by the artificial heart team at the University of Virginia, are as follows:

1) to produce adequate cardiac output to ensure tissue perfusion;
2) to support native cardiac function (i.e., antegrade flow across aortic valve);
3) to adjust cardiac output to compensate for increased physiological need due to activity;
4) to avoid left ventricular collapse due to mechanical suction from the pump;
5) to provide sufficient pump speed to avoid retrograde pump flow from aortic-left ventricular pressure differential.

All five objectives of LVADs are related to either blood pressure or flow rate, among which, 1), 3), and 4) are critical, while 2) and 5) are secondary. Objective 2) is proposed because of the concern of blood coagulation on the aorta side of aortic valve. It may contradict objective 1) in some case, and should be ignored in the controller design if so, but be compensated by proper anticoagulation drug treatment. Objective 5) is highly preferred to prevent recirculation of blood and the accompanying thrombosis factors, but not a must. Expressing in terms of hemodynamic variables and pump variables, these objectives mean: TPF should be maintained at 5–6 L/min in rest for an average person, and be varied properly in different activities or pathologic states to match the varied metabolism (maybe up to 2–3 times); $LVPED$ should be maintained between two limits (for example -3 mmHg to 15 mmHg); LVAD flow rate should be maintained always positive if possible [4]. The upper limit of $LVPED$ is recommended to relieve the pulmonary congestion, and is allowed to be crossed for a short period occasionally. The lower limit of $LVPED$ is time critical since it is proposed to prevent left ventricular collapse, which may result in severe damage to the heart muscle, even the death of LVAD recipient.

To regulate these hemodynamic variables, they need to be detected first by corresponding sensors. In artificial heart pumps, long-term physiological sensors for hemodynamic variables (blood flow, pressure) are not available, due to the constriction of volume, blood compatibility, and reliability issues. Long term available engineering signals, such as pump head and motor signals, do not have direct physiological meanings. Therefore, to meet these objectives, a state estimator, which can derive the hemodynamic variable (blood flow or pressure) from available feedback signals, is needed for the design and implementation of a physiological control system for a long-term LVAD.

IV. Review of Control Designs

Many rotary LVADs are operated at a constant speed, such as the Jarvik 2000 [6] and the DeBakey pump. The DeBakey pump also allows the physician or trained personnel to manually adjust the pump speed until a perceived comfort level of perfusion is achieved. These constant speed settings or manual adjustment

are adequate for short period usage of LVADs, but are likely to be insufficient for long-term applications. The current design of the physiological control system, which automatically responds to various real-time physiological demands, is discussed here.

1) Suction Limit Control: To ensure enough perfusion for the body, many CF LVADs are always operated at the highest operating speed as determined by when the left ventricular collapse does not happen [15]–[20], [27]. Many control algorithms rely on the motor current waveform to detect the left ventricular suction. When suction occurs, the motor current amplitude [15], the harmonics of motor current [16]–[18], the shape of motor current waveform [19], and the ratio of motor current in systole to its value in diastole [5], will reach minimum or show abnormality. Chen *et al.* suggested that the gradient of LVAD flow in diastole with respect to pump speed reached minimum when suction happened [27]. These algorithms are pertinent to a specific pump, and may cause overperfusion due to the difficulty to detect the overperfusion/danger-of-suction region with suction limit algorithms [19]. Overperfusion may result in the dehydration of patients.

2) P/PI Control: Parnis *et al.* used a proportional (P) controller for the Jarvik 2000 VAD. The pump rotational speed was set as a linear function to the heart rate, which was obtained from the fundamental frequency of the motor current waveform [7]. Treadmill exercise and heart pacing studies were performed on a calf with the Jarvik 2000 VAD and no deleterious effects were detected within test durations in excess of 26 weeks [7]. However, a linear function between the heart rate and pump rotational speed has an obvious discrepancy. It has been found that the cardiac output of a healthy subject is not only determined by the heart rate but also by the contractability of the heart, and most importantly determined by the peripheral circulation [8]. Waters *et al.* designed a PI controller for an LVAD using one feedback signal, the pump head of LVAD [9]. The beating of the natural left ventricle was oversimplified as a sinusoidal disturbance to the LVAD.

3) Fuzzy Logic Control: A fuzzy controller was employed to regulate the LVAD flow to track a desired flow rate [10]. The LVAD flow rate was estimated from the motor current and rotational speed. The desired LVAD flow rate was assumed proportional to the heart rate. This assumption ignored the influence of heart contractility, and most importantly, the effect of the peripheral circulation on the desired flow rate [8]. Choi *et al.* also implemented a PI-type fuzzy controller to realize an LVAD pump flow rate pulsatility tracking [11]. A reference pulsatility level of 15 mL/s was selected to allow the natural heart to produce some stroke volume without introducing ventricular suction. The LVAD pump flow rate was estimated from the motor current and rotational speed. Simulation results showed that the control signal of this fuzzy controller produced a much smaller variation in the LVAD speed with regard to parameter variations than that of a PI controller. However, the constant setting of the reference pulsatility may be inadequate since the failure level of the natural ventricle during the long period of LVAD support may change, resulting in different abilities to generate flow rate pulsatility.

4) Optimal Control: This type of controller was designed to minimize some predefined penalty functions. An optimal controller with the structure of a PI controller was designed, whose time varying coefficients were obtained from offline numerical searches to minimize a weighting function [12], [28]. This function was the combination of pump head or differential pressure between the pulmonary vein and the aorta and the variation rate of the rotational speed. When the physical cardiovascular condition is different from the conditions used in the numerical searches, the optimality of this controller may not be guaranteed. A similar but more complex optimal controller was designed by Boston *et al.* to minimize a multiobjective penalty function $J(\omega) = \sum_{i=1}^{3} \mu_i J_i(\omega)$ [13], where $J_i(\omega)$ was the penalty function of cardiac output, arterial pressure, and left atrial pressure, respectively, and $\mu_i \geq 0$ with $\sum_{i=1}^{3} \mu_i = 1$. The LVAD motor speed was used as the feedback signal. This technique required simple and accurate predetermined mathematical models of cardiac output, arterial pressure, and left atrial pressure with respect to the motor speed.

Briefly speaking, many current controllers for rotary LVAD only addressed the issue to prevent the collapse of the left ventricle [5], [15]–[20], [27]. Some controllers regulated nonphysiological signals directly [7], [9]–[12], whose relations with the human circulation system need further exploration [11], [13]. Some controllers need the measurement of hemodynamics variables, which cannot be reliable by present sensors in the long-term [27], [28]. Most controllers are not studied in possible physiological conditions that may occur in the lifetime of a permanent LVAD, such as activity or pathological variations. Therefore, an effective and robust control system, which addresses the issues of sensors, estimator, controller, prevention of left ventricular suction, and robust performance in the presence of physiological variations, is in demand for a permanent rotary LVAD.

V. Simplified System Model

The LVAD is connected to the left ventricle (pump inlet) and the aorta (pump outlet). Therefore, from the LVAD's point of view, only the hemodynamic response of the left ventricle and the aorta is pertinent. To develop an effective robust and adaptive controller to meet the desired physiological need, we consider a simplified cardiovascular system model shown in Fig. 2 (as compared with the 11th-order model shown in Fig. 1). Pulmonary circulation (blocks between P_4 and P_7 in Fig. 1, including C_7) is simplified as a pressure resource δ_P. Simulation results of the 11th-order model demonstrated that δ_P hardly vary over time, with the mean value less than 5 mmHg [4]. C_2 and C_3 in Fig. 1 are combined together as C_A in Fig. 2, representing the capacitance of the systemic artery. C_o, C_m, and C_4 in Fig. 1 are combined as C_V in Fig. 2, representing the capacitance of the systemic vein and the right atrium. The resistors between P_2 and P_4 in Fig. 1 are represented by one variable resistor TPR in Fig. 2. The muscle pump function (pressure resource P_{muscle} together with diodes D_{m_1} and D_{m_2} in Fig. 1) is simplified as a pressure resource δ_m, whose mean value is found by simulation results to be less than 5 mmHg. The simulation results also demonstrated that the response of the simplified model in the left ventricle and the aorta is very close to the response at these two locations of the 11th-order model in

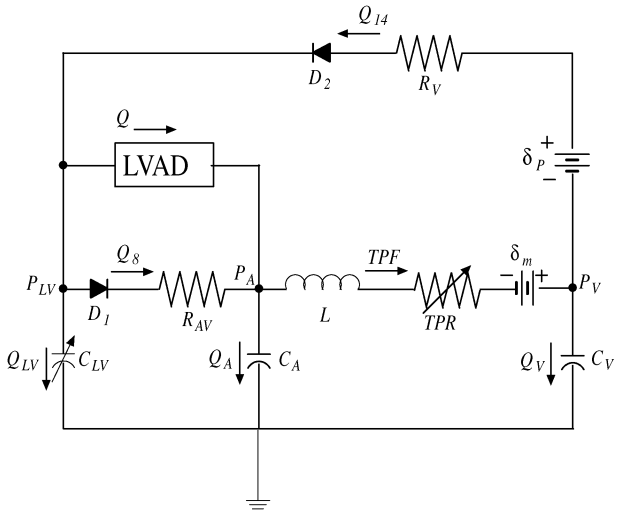


Fig. 2. Simplified system model.

Fig. 1, thus, they provide enough accurate information for the LVAD controller design [4].

In this model, the system elements are: $C_{\rm LV}$ left ventricle; C_A aorta; C_V systemic vein and right atrium; D_1 aortic valve; D_2 mitral valve; TPR total peripheral resistance; L blood inertia in peripheral circulation; R_{AV} resistance of aortic valve; R_V resistance of pulmonary vein and left atrium; δ_m the pressure disturbance generated by the muscle pump function in exercise; δ_P pressure disturbance generated by pulmonary circulation; and the system signals are: $P_{\rm LV}$ left ventricular pressure; P_A aortic pressure; P_V central venous pressure; $Q_{\rm LV}$ inflow to left ventricle; Q_{14} flow rate from the vein to the left ventricle; Q pump flow rate; Q_8 flow rate from the left ventricle to the aorta; Q_A inflow to aorta; Q_V inflow to systemic vein and right atrium; TPF total peripheral flow rate. The values of the parameters in Fig. 2 are listed in the Appendix.

TPF and $LVPED$, discussed in Section III, are related to the mean aortic pressure $\bar{P}_A$ [8], as $\bar{P}_A \approx TPR * TPF$ and $LVPED$ depend on TPF. Therefore, the objectives of physiological control can be reexpressed as the requirement for average aortic pressure [4], which is a continuous signal. The simplified model developed in this section will be used to derive the aortic pressure using available feedback signals.

For a CHF patient with LVAD, in diastole, D_2 is open and D_1 is closed. In systole, D_2 is closed and D_1 may be open depending on whether $P_{\rm LV} > P_A$ is true or not. As in [4], the dynamic equations of this system model are

$$\begin{aligned} \dot{x} &= A^d x + B_1^d Q + B_2\delta_P + B_3\delta_m \\ y &= Cx \text{ in diastole} \end{aligned} \tag{5.1}$$

$$\begin{aligned} \dot{x} &= A^s x + B_1^s Q_s + B_3\delta_m + B_4\delta_{\rm LV} \\ y &= Cx \text{ in systole} \end{aligned} \tag{5.2}$$

where the superscripts "s" and "d" indicate the matrices during the systole and the diastole, respectively, $x = [x_1, x_2, x_3]^T$, $x_1 = P_{\rm LV} - P_A, x_2 = P_V - P_A, x_3 = TPF, Q_s = Q + Q_8$, $y = \Delta P = P_A - P_{\rm LV}$

$$A^d = \begin{bmatrix} -\frac{1}{C^d_{\rm LV}R_V} & \frac{1}{C^d_{\rm LV}R_V} & \frac{1}{C_A} \\ \frac{1}{C_V R_V} & -\frac{1}{C_V R_V} & \frac{1}{C_A}+\frac{1}{C_V} \\ 0 & -\frac{1}{L} & -\frac{TPR}{L} \end{bmatrix}$$

$$B_1^d = \begin{bmatrix} -\frac{1}{C^d_{\rm LV}} - \frac{1}{C_A} \\ -\frac{1}{C_A} \\ 0 \end{bmatrix}$$

$$B_2 = \begin{bmatrix} \frac{1}{C^d_{\rm LV}R_V} \\ -\frac{1}{C_V R_V} \\ 0 \end{bmatrix}$$

$$B_3 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}$$

$$A^s = \begin{bmatrix} 0 & 0 & \frac{1}{C_A} \\ 0 & 0 & \frac{1}{C_A}+\frac{1}{C_V} \\ 0 & -\frac{1}{L} & -\frac{TPR}{L} \end{bmatrix}$$

$$B_1^s = \begin{bmatrix} -\frac{1}{C^s_{\rm LV}(t)} - \frac{1}{C_A} \\ -\frac{1}{C_A} \\ 0 \end{bmatrix}$$

$$B_3 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}$$

$$B_4 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\delta_{\rm LV} = \frac{V_{\rm LV}}{\left(C^s_{\rm LV}(t)\right)^2}\dot{C}^s_{\rm LV}(t)$$

$C = [-1, 0, 0]$, and $Q_8 = (\Pi(x_1)/R_{AV})$. The value of the unit step function $\Pi(x_1)$ for an LVAD recipient is zero for the majority of the heartbeat [4]. This state–space model converts the variation of the left ventricular capacitance in systole to an extra input $\delta_{\rm LV}$.

In this model, $C^d_{\rm LV}$ is a constant obtained by linearizing the exponential relation of pressure and volume in diastole, which shows a strong linear relation over the range of volume for a LVAD recipient [4]. $C^s_{\rm LV}(t)$ is the capacitance of the left ventricle in systole. The output y is the measured pump head of LVAD, that can be obtained from long-term reliable engineering sensors of the magnetically levitated LVAD under development [4]. The control input Q is the pump flow rate which can be calculated from the pump characteristic equation [4]

$$\frac{Q}{\omega} = \alpha_2\left(\frac{\Delta P}{\omega^2}\right)^2 + \alpha_1\frac{\Delta P}{\omega^2} + \alpha_0 \tag{5.3}$$

for the pump speed ω and some constants α_2, α_1, and α_0.

The key parameter in (5.1) and (5.2) is TPR which changes as the activity level changes. This parameter needs to be estimated for the design of a state observer for the system state x, and for the design of a feedback controller to make the aortic pressure P_A to follow a desired trajectory. It is important to note

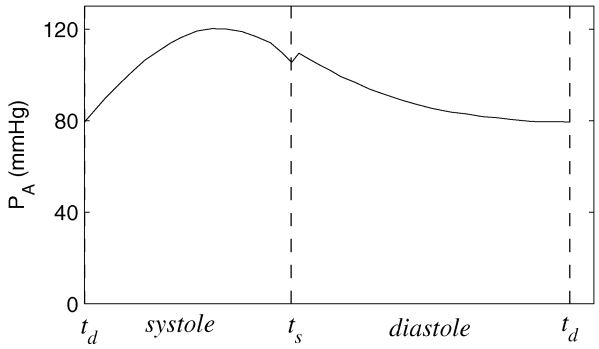


Fig. 3. Waveform of aortic pressure during one heart cycle.

that a state observer is needed, because the aortic pressure P_A is not available for measurement. The blood volume equation

$$P_{\rm LV}C_{\rm LV}+P_AC_A+P_VC_V=V_s=\text{total stressedblood volume} \tag{5.4}$$

could be used to calculate the aortic pressure P_A if x_2 and x_3 were available (note that $x_1=-y$ is available)

$$P_A=\frac{[-C_{\rm LV},-C_V,0]x+V_s}{C_{\rm LV}+C_A+C_V}\triangleq Hx+N. \tag{5.5}$$

1) Stability Analysis: For this system model, the system matrix switches from A^d to A^s, when the system state changes from diastole to systole. We have shown that the homogeneous part ($\dot{x}=Ax$) of this system is Lyapunov stable at $x_e=[0,0,0]^T$, by finding a common Lyapunov function: $V=x^TMx$ for a constant symmetric and positive definite matrix, such that $\dot{V}<0$ for $\dot{x}=A^dx$ (diastole) and $\dot{V}\le 0$ for $\dot{x}=A^sx$ (systole) [4] (it can be further shown to be asymptotically stable [4], and thus exponentially stable).

The response of the system (5.1), (5.2), in the steady state, is a stable oscillation. The typical waveform of the aortic pressure P_A over the time period of one heartbeat is shown in Fig. 3, where the time interval for diastole is $[t_s,t_d)$ and the time interval for systole is $[t_d,t_s)$.

With the ensured stability, the system (5.1), (5.2) can be approximated by a single-equation model

$$\begin{aligned}\dot{x'}&=Ax'+B_1Q_s+B_2'\delta_P+B_3\delta_m\\ y'&=Cx'\\ A&=\begin{bmatrix}-\frac{1}{C^d_{\rm LV}R'_V} & \frac{1}{C^d_{\rm LV}R'_V} & \frac{1}{C_A}\\ \frac{1}{C_VR'_V} & -\frac{1}{C_VR'_V} & \frac{1}{C_A}+\frac{1}{C_V}\\ 0 & -\frac{1}{L} & -\frac{TPR}{L}\end{bmatrix}\\ B_2'&=\begin{bmatrix}\frac{1}{C^d_{\rm LV}R'_V}\\ -\frac{1}{C_VR'_V}\\ 0\end{bmatrix}\end{aligned} \tag{5.6}$$

and $B_1=B_1^d$, where R'_V is the equivalent venous resistance, an artificial parameter. The value of R'_V can be determined as

$$R'_V=\frac{x'_2+\delta_P-x'_1}{Q_s}=\frac{x_2+\delta_P-x_1}{Q_s}\Big|_{t=t_s}\quad\text{steady state} \tag{5.7}$$

by the measurements in LVAD implantation surgery before switched to physiological control operation mode. (5.7) is derived from the steady-state solution of (5.6) because during surgery, the LVAD usually runs at constant speed. The response of human circulatory system thus follows the steady-state solution of (5.6).

In particular, the circulatory system (5.1), (5.2) and the system (5.6) was shown to have the same solution at $t=t_s$ using the state transition matrix over one heartbeat, assuming that Q_s and δ_m are constant during systole and during diastole [4]. The output of (5.6) y' at $t=t_s$ is equal to the output of (5.1), (5.2), i.e., ΔP, which reaches the maximum in each heart cycle at $t=t_s$. For the LEV VAD, ΔP is derivable from the axial movement of the impeller. With ΔP, discrete signal $\Delta P|_{t=t_s}=(\Delta P)_{\max}$ in each heart cycle can be obtained. Passing this discrete signal through a zero-hold function and a low-pass filter, continuous signal y' can be generated and used as the feedback signal. It is worth noting that the value of R'_V in system (5.6) is much bigger than the true venous resistance R_V in (5.1), implying a slower converging speed of (5.6) than (5.1).

System (5.6) provides a linear system whose parameter estimator and state estimators are easy to design. Since states of human circulatory system (5.1), (5.2) are equal to the states of system (5.6) at $t=t_s$, we can obtain the states of the human circulatory system at t_s in each heart cycle from the states of (5.6). Furthermore, the average value of the aortic pressure, which is determined more by the bottom envelope of states than by the instantaneous value of states, is of more interest in controlling LVAD than the instantaneous value of aortic pressure. The bottom envelope of the states of system (5.1), (5.2) can thus be approximated with the state estimate of system (5.6) in the whole heart cycle besides at $t=t_s$, and be used in (5.5) to determine the average aortic pressure in each heart cycle.

With the value for R'_V in system (5.6) found accurately, the approximation error can be very small. Studies in [31] demonstrated that big fluctuation of input, variation of systole versus diastole ratio and heart rate will increase the approximation error because the perfectly matched value for R'_V may change in these conditions. However, for each specific LVAD recipient, variation of these parameters in different physiological states is much smaller than the studied range, resulting in negligible variation of the value for R'_V [31].

Based on this single-equation description of the circulatory system, a parameter estimator, a state observer, and a feedback controller, can be developed for physiological control. From this system expression (the presence of the parameter $C^d_{\rm LV}$), we see that the diastolic behavior is more dominant in the circulatory system response in a mean sense. The disturbance caused by $\delta_{\rm LV}$ in (5.2) does not show in (5.6).

VI. Adaptive Optimal Controller

In this section, we present the detailed results of our adaptive optimal physiological controller whose design consists of three parts: an adaptive parameter estimation scheme to estimate the TPR, an adaptive state observer using the estimate of TPR, and an optimal PI controller also using the estimate of TPR for controller parameter selection. This physiological control system is shown in Fig. 4.

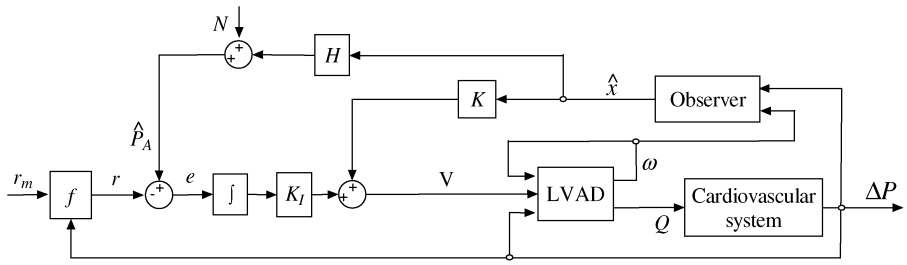


Fig. 4. Physiological control system structure. r: reference pressure of aortic pressure. r_m: preset constant reference pressure. ΔP: pump head of the LVAD (human cardiovascular system output). e: tracking error of aortic pressure to r. $\hat{x}$: the estimated state of system x. $\hat{P}_A$: the estimated aortic pressure. H: transformation matrix from x to $\hat{P}_A$. K_I. K: PI controller gains. V: the control output (the desired pump motor voltage).

1) Parameter Estimation: From (5.6), it follows that:

$$(s^3+a_2s^2+a_1s+a_0)y'(s)=(b_2s^2+b_1s+b_0)Q_s(s) +\left(b_2^Ps^2+b_1^Ps+b_0^P\right)\delta_P(s)+\left(b_2^ms^2+b_1^ms+b_0^m\right)\delta_m(s) \quad (6.1)$$

for some constants a_i and b_i (which depend on TPR), b_i^P and b_i^m, $i = 0, 1, 2$. The nominal value of δ_P for LVAD recipient can be measured in surgery, which is usually less than 5 mmHg [4]. The change of δ_P in the variation of activities is negligible. The average value of δ_m was shown in simulation to be less than 5 mmHg, less than 5% of Q_s and y' [4]. In the parameter estimation of TPR, δ_m is ignored since it is not accessible and only effective in exercise. The corresponding estimation error will be 5 mmHg at most, and can be compensated as discussed at the end of this section.

Filtering both sides of (6.1) by a low-pass filter $F(s) = (1/\Lambda(s))$ with $\Lambda(s) = s^3 + 15s^2 + 75s + 125$, and arranging the resulting terms, we can derive the parametric equation

$$\theta^*\phi(s) = z(s) \quad (6.2)$$

for $\theta^* = TPR$ and some functions $\phi(s)$ and $z(s)$ whose denominators are $\Lambda(s) = s^3 + 15s^2 + 75s + 125$ and numerators are combinations of $y'(s)$, $Q_s(s)$, and $\delta_P(s)$ (with δ_m ignored). The parameters of $\Lambda(s)$ with the cutoff frequency of 5 rad/s (about 48 beats per minute, minimal heart rate expected for LVAD recipient) were selected to eliminate the disturbance introduced by the heartbeat and the high frequency components of the muscle pump function.

In the time-domain, we define the estimation error $\epsilon(t) = \theta(t)\phi(t) - z(t)$, where $\theta(t)$ is the estimator of θ^*, and use the following adaptive law to update the estimate $\theta(t)$:

$$\dot{\theta}(t) = -\frac{\gamma\epsilon(t)\phi(t)}{\sqrt{1+\phi^2(t)}}, \qquad \gamma > 0. \quad (6.3)$$

In terms of the unknown parameter error $\tilde{\theta}(t) = \theta(t) - \theta^*$, we can express this adaptive law as $\dot{\tilde{\theta}}(t) = -(\gamma\phi^2(t)/\sqrt{1+\phi^2(t)})\tilde{\theta}(t)$. In this case, when $\theta(t)$ is a scalar parameter, the error $\tilde{\theta}(t)$ converges to zero, that is, $\lim_{t\to\infty}\tilde{\theta}(t) = 0$, if $\int_0^\infty -(\gamma\phi^2(t)/\sqrt{1+\phi^2(t)})dt = \infty$ (it is satisfied if the signal $\phi(t)$ does not vanish). In practice, due to modeling errors, parameter variations, and disturbances, this ideal property may only be met approximately. In our application, the parameter θ^* is varying between two values (from exercise to rest, or vice verse), the adaptive law is to provide an online estimate of this parameter.

2) State Observer: The state observer structure for the estimate $\hat{x}$ of $x = [x_1, x_2, x_3]^T$ in (5.1), (5.2) is a standard one, based on the equivalent model (5.6) with all parameters but TPR known and δ_m ignored. The adaptive estimate of TPR obtained online from the previous parameter estimation procedure is used, leading to an adaptive observer. The estimate of P_A is given as $\hat{P}_A = H\hat{x} + N$ [see (5.5)].

3) Optimal PI Controller: The objective of physiological control is to raise the aortic pressure P_A to a certain level (95 mmHg as the nominal value). This objective is achieved by a feedback control design and a chosen reference signal $r(t)$ through a signal generator f (see Fig. 4).

The motor equation of an LVAD can be described by

$$J\dot{\omega} = K_tI - T_{\text{load}}, \qquad L\dot{I} + RI + K_e\omega = V \quad (6.4)$$

where V is the applied voltage, ω is the pump rotational speed, I is the motor current, and $T_{\text{load}} = (\Delta PQ/\omega\eta)$ is the blood hydraulic torque on the pump impeller, with η being the pump efficiency. Since the motor inductance L and the pump moment of inertia J are small, the motor equation is simplified as

$$V = \frac{\Delta PQR}{\omega\eta K_t} + K_e\omega. \quad (6.5)$$

This relationship, together with that in (5.3), gives an expression of Q in terms of ΔP and V, which can be linearized as $Q = \beta_1\Delta P + \beta_2 V$ for some linear coefficients β_1 and β_2 [4]. The implication of such an expression is that the (5.6) can be reexpressed with V as the input signal [4]. The control objective now is to find a feedback control signal (voltage) V for the reexpressed (5.6) such that the aortic pressure P_A (through its estimate $\hat{P}_A$ from the adaptive state observer) tracks a reference signal $r(t)$ chosen to meet certain physiological needs.

For the control system shown in Fig. 4, K_I is an integral gain, K is a feedback gain vector, both calculated from an optimal control design procedure to minimize the energy of a weighted function of control output and tracking error, given as $W = \int(e^2S + V^2R)dt$ [4]. S and R are the selected weighting factors for the aortic pressure tracking error $e(= r - \hat{P}_A)$ and control voltage V, respectively. The estimate of TPR is used to replace the unknown TPR in optimal control and state observation, leading to an adaptive control scheme [4]. To derive

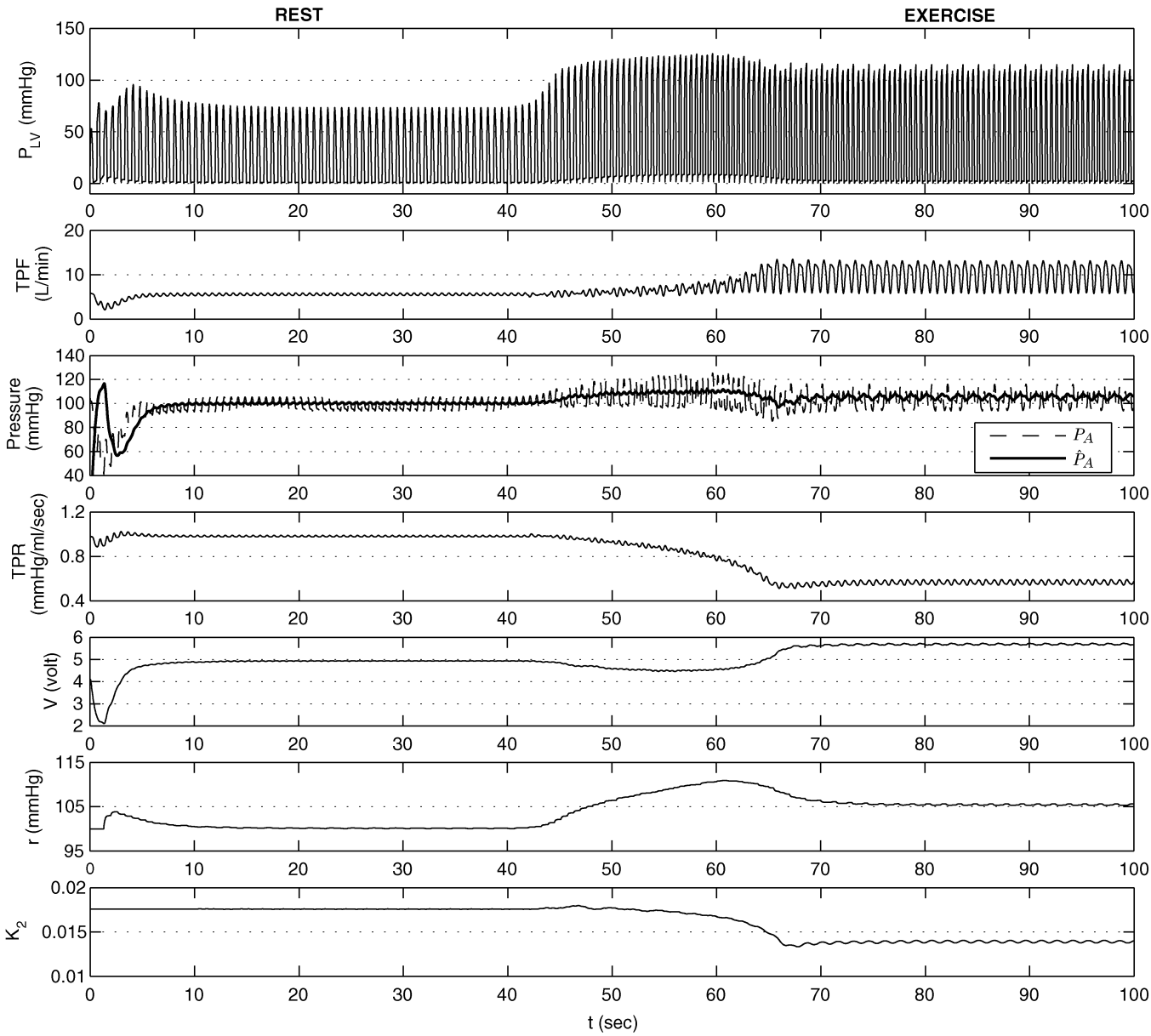


Fig. 5. Simulation results. $P_{\rm LV}$: left ventricular pressure. P_A: aortic pressure. TPF: total peripheral flow. P_A (dashed line): aortic pressure. $\hat{P}_A$ (solid line): estimated aortic pressure. TPR: total peripheral resistance. V: control input voltage. r: reference value. K_2: the second element of controller gain.

such a control scheme, we started with the linearized expression $Q = \beta_1 \Delta P + \beta_2 V$, to express the system (5.6) as

$$\begin{aligned} \dot{x'} &= (A + B_1\beta_1 C)x' + B_1\beta_2 V + B_1 Q_8 + B_2'\delta_P + B_3\delta_m \\ \Delta P' &= Cx'. \end{aligned} \tag{6.6}$$

An adaptive state observer using system parameters and an adaptive estimate of TPR is constructed to generate an estimate $\hat{x}$ of x and an estimate $\hat{P}_A$ of P_A. The adaptive optimal controller uses $\hat{P}_A$ for feedback control.

4) Reference Signal $r(t)$: In this physiological control scheme, the reference signal $r(t)$ is generated from a design function f and an auxiliary signal r_m, using a precompensation algorithm: $r(t) = r_m + f(\Delta P)$. r_m is a preset constant equal to the normal value of mean arterial pressure, i.e., 95 mmHg for an average person. The choice of f is not unique, if only f is inversely correlated to ΔP. This relation is based on the observation that: if the LVAD flow rate Q (left ventricle outflow) is lower than the venous return Q_{14} (left ventricle inflow), which is an indication of the human body's need, the left ventricular pressure $P_{\rm LV}$ increased, and in turn decreased ΔP. If Q is higher than Q_{14}, ΔP increased. In this paper, the function f is a dead zone function of ΔP as

$$f(\Delta P) = \begin{cases} 0, & \text{for } |\Delta P - r_{\rm diff}| \le a \quad (6.7a) \\ \lambda(\Delta P - r_{\rm diff}), & \text{for } |\Delta P - r_{\rm diff}| > a \quad (6.7b) \end{cases}$$

for some design constants $\lambda < 0$, $r_{\rm diff} > 0$, and $a > 0$. $r_{\rm diff}$ is the dead zone limit (30 mmHg in this paper), set to avoid the unnecessary update of $r(t)$ within each heart cycle due to the heartbeat. The value of a (equal to 1 in this paper) should be adjusted for each specific patient so that the combination of a and $r_{\rm diff}$ results in a reference signal r equal to r_m in average when the patient is perfused properly.

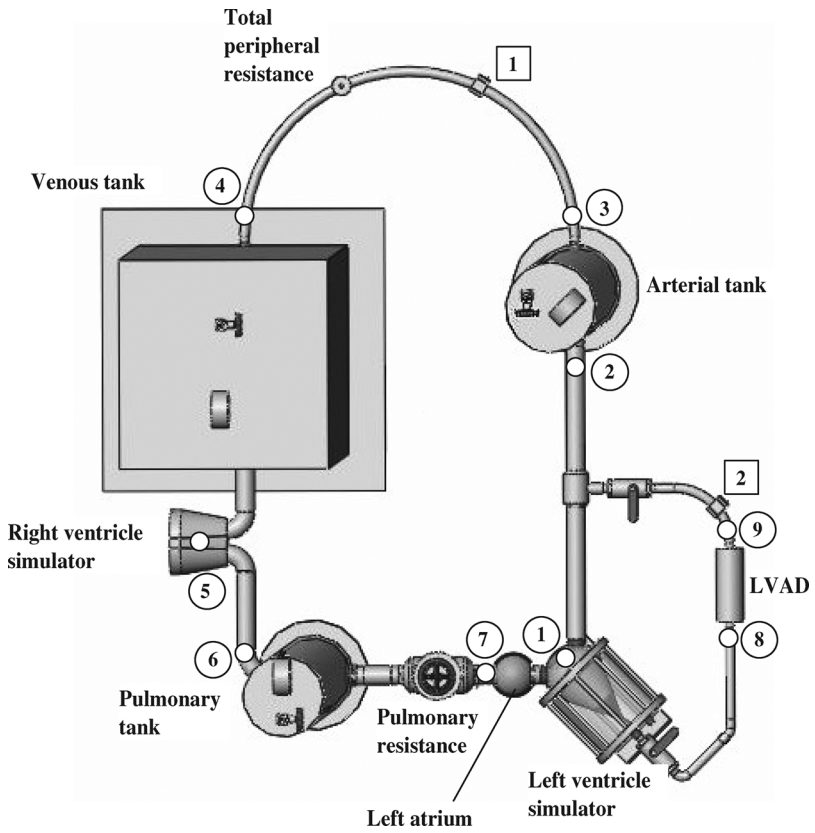


Fig. 6. Mock human circulatory loop [14].

The update of $r(t)$ is important because the body's need for blood flow may vary a lot in the presence of physiological state variation. In these variations, some parameters of the human circulatory system, such as heart rate, TPR, ventricular elastance $(E_{\max})$, may change significantly, leading to the variation of the venous return and left ventricular pressure $P_{\rm LV}$. As a result, the pump head ΔP will exhibit a variation too [4]. The update of reference signal $r(t)$ by the designed f function of ΔP can guarantee that the body's need can always be matched by the LVAD flow rate, and prevent the underperfusion and overperfusion of the human body that may happen if constant reference value is used in the variation of physiological states. Also, the update of $r(t)$ by $f(\Delta P)$ can compensate for the estimation error due to the ignorance of δ_m. In exercise, the estimation error of aortic pressure will be increased because of the ignorance of δ_m. Supposing $r(t)$ is the same, the actual aortic pressure will be elevated. Meanwhile, however, ΔP increases, leading to the decrease of $r(t)$ by $f(\Delta P)$ in (6.7). So, the true aortic pressure will actually be maintained in a level ensuring proper perfusion of patients.

VII. Simulation Study

The control system shown in Fig. 4 was simulated using Matlab/Simulink, with the controller designed based on the reduced-order model (5.6) and tested in the full-order system shown in Fig. 1. Extensive simulation results for various healthy and CHF cases have been obtained, which validated our analytical work in modeling, estimation, and the control of the human cardiovascular circulatory systems with a LVAD [4].

Fig. 5 illustrates the simulation results of the 11th-order model (Fig. 1) from rest to exercise with the LVAD and its physiological controller. TPR decreases dramatically (about 50%) in exercise, and TPF increases significantly. Heart rate and the contractility of the left ventricle are increased in exercise, and $P_{\rm muscle}$ exhibits a nonzero sinusoidal-like waveform. Before 10 s, the value of r is set to be 100 mmHg. After 10 s, the value of r is determined by $r(t) = r_m + f(\Delta P)$. The average estimation error for aortic pressure is maintained less than 1 mmHg in this activity variation. The value for reference signal r changes from 100 mmHg in rest to 105.4 mmHg in exercise. TPF, $LVPED$, and the average aortic pressure $(\bar{P}_A)$, are restored by the LVAD to: TPF: 5.6 L/min in rest and 9.8 L/min in exercise; $LVPED$: 1.1 mmHg in rest and 2.3 mmHg in exercise; and $\bar{P}_A$: 99.6 mmHg in rest and 103.2 mmHg in exercise, respectively. These values are close to the corresponding simulation results for a healthy person, i.e., TPF: 5 L/min in rest and 9.9 L/min in exercise; $LVPED$: 10.2 mmHg in rest and 12.4 mmHg in exercise; $\bar{P}_A$: 97.1 mmHg in rest and 102 mmHg in exercise, respectively. As a comparison, these values for a CHF patient without LVAD are: TPF: 2.9 L/min in rest and 5.5 L/min in exercise; $LVPED$: 27.8 mmHg in rest and 33.6 mmHg in exercise; and $\bar{P}_A$: 97.8 mmHg in rest and 103.6 mmHg in exercise. The value of TPR for a CHF patient without LVAD is set to a higher value than that of a CHF patient with LVAD to match the vasorestriction effect of the human body in the presence of blood flow drop.

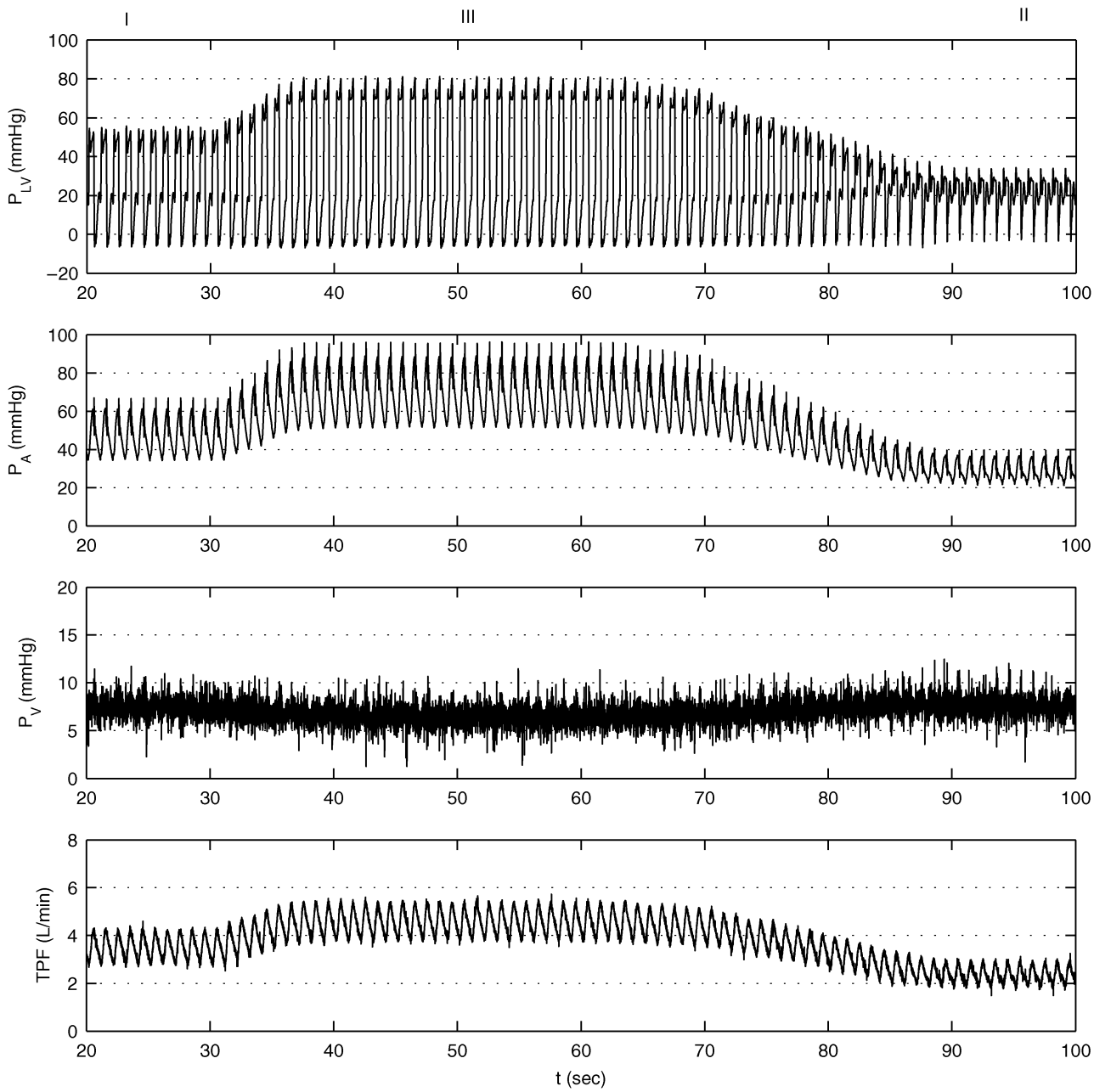


Fig. 7. Experimental results of pathological state variations (without MY2 pump). $P_{\rm LV}$: left ventricular pressure. P_A: aortic pressure. P_V: venous pressure. TPF: total peripheral flow.

VIII. Experimental Study

A mock human circulatory loop was set up as an in vitro test rig for different versions of prototype LVADs, as shown in Fig. 6. The components of the loop mimic the key components of the human cardiovascular system. This test rig can simulate different normal or pathologic states and activities of a cardiovascular system [14]. A small centrifugal pump MY2 (Speck Pump Inc., Jacksonville, FL) was used in the place of an LVAD in the testing.

Study of Pathologic State Variations: Figs. 7 and 8 illustrate the experimental results of a mock human loop in different pathologic states without and with the MY2 pump, respectively. Three different pathologic states in rest condition, namely I, II, and III are reproduced exactly the same in Figs. 7 and 8. The systolic function of the left ventricle simulator is depressed in these states, among which II is the worst, III is the best. MY2 pump is controlled by a real-time controller board DS 1104 (dSPACE Inc., MA). The estimated aortic pressure $\hat{P}_A$ shows delay from the true aortic pressure P_A by approximately one heart cycle due to the zero-hold function and low-pass filter in obtaining y' in (5.6) from $\Delta P|_{t=t_s} = (\Delta P)_{\max}$. The average estimation error for aortic pressure is maintained less than 2 mmHg. The abnormal hemodynamic variables, such as TPF, $LVPED$, and the average aortic pressure $\bar{P}_A$, are all restored to the normal physiological range, 5–6 L/min, $<$15 mmHg, and $\sim$95 mmHg, respectively, by the designed physiological controller in the presence of pathologic state variations. As a com-

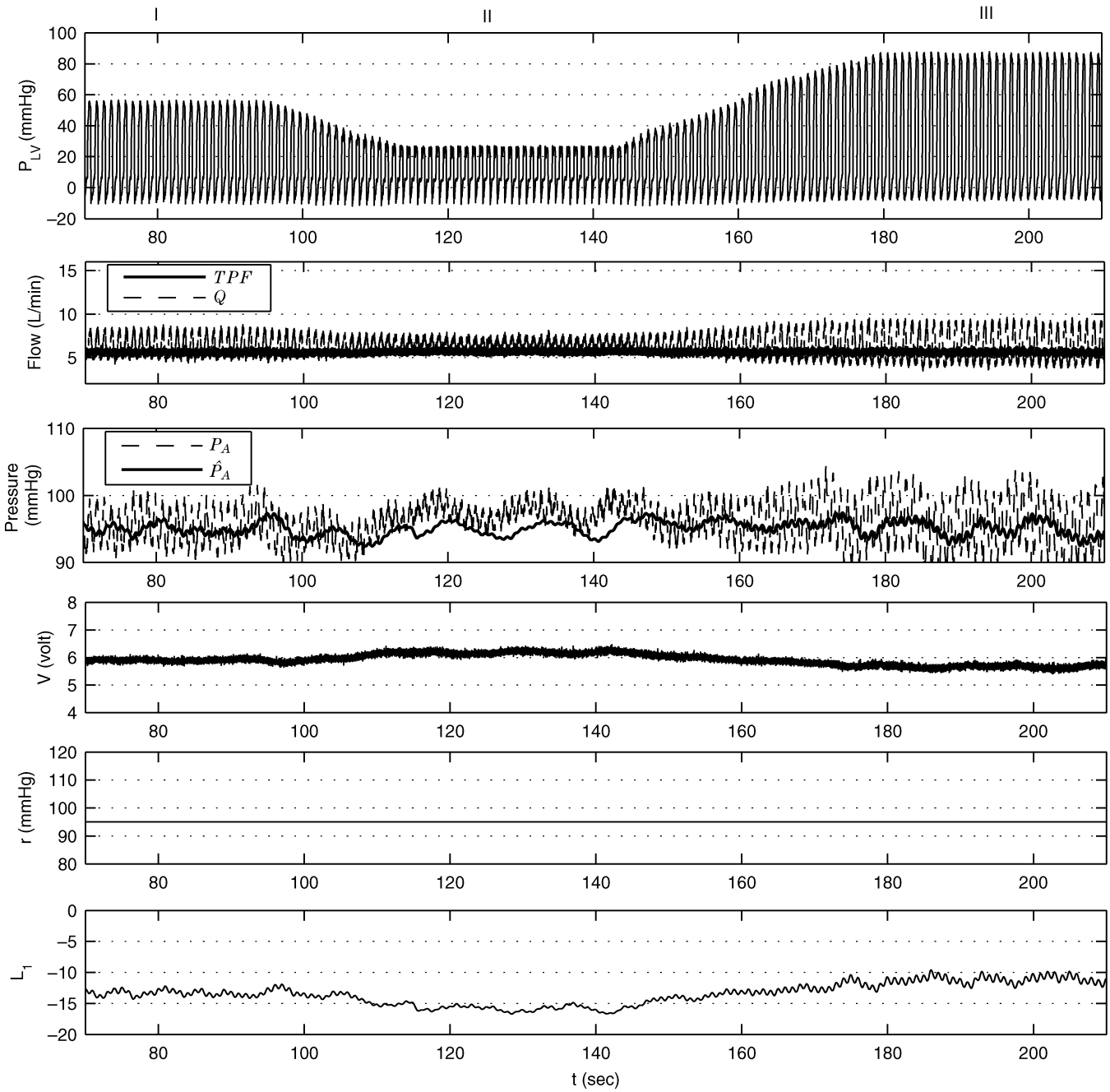


Fig. 8. Experimental results of pathological state variations (with MY2 pump). $P_{\rm LV}$: left ventricular pressure. TPF: total peripheral flow. Q: LVAD flow rate. P_A (dashed line): aortic pressure. $\hat{P}_A$ (solid line): estimated aortic pressure. V: control input voltage. r: reference value, L_1: the first element of observer gain.

parison, these values for a CHF patient without LVAD are as follows: TPF: 3.5 L/min in state I, 2.3 L/min in state II: 4.6 L/min in state III; $LVPED$: 16.2 mmHg in state I, 17.4 mmHg in state II, and 20 mmHg in state III; $\bar{P}_A$: 47.2 mmHg in state I, 28.8 mmHg in state II, and 68.7 mmHg in state III. The reference signal r is set constant with the value of 95 mmHg without an online update in the experiment because the left ventricular simulator was unable to reproduce the relation between the left ventricular pressure $P_{\rm LV}$ and the venous return [4], thus invalidate the use of the f function to update $r(t)$ derived upon that relation.

Study of Suction Reverse: One potential problem of rotary LVADs is the collapse of left ventricle due to an improperly high pump speed. To test if the designed physiological controller can reverse suction, the scenario of left ventricular collapse was simulated in the mock circulatory loop by setting the operational speed of the MY2 pump at 4050 r/min with pathologic level I in the rest condition. Then the pump was switched to the operating mode of the designed physiological controller. During the whole test, none of the components in the mock circulatory loop was changed. The experiment results of this case are shown in Fig. 9. In the suction condition, the $LVPED$ was approximately -10.2 mmHg. Once the pump was switched to the working mode of the physiological controller, the $LVPED$ increased, even higher than 15 mmHg for a short period (approximately 8 heart cycles), then decreased again and approached

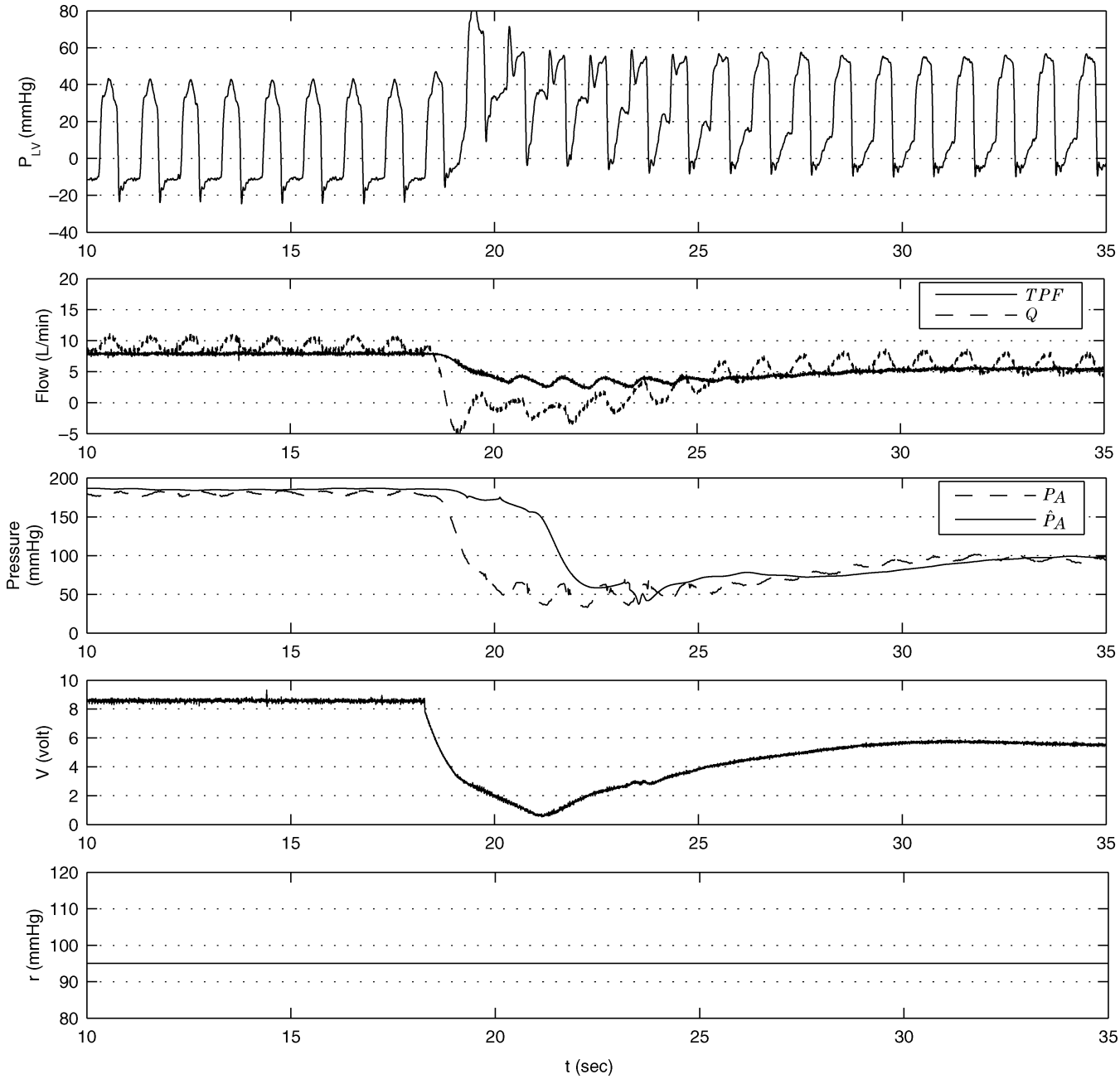


Fig. 9. Experimental results of suction reverse (with MY2 pump). $P_{\rm LV}$: left ventricular pressure. TPF: total peripheral flow. Q: LVAD flow rate. P_A (dashed line): aortic pressure. $\hat{P}_A$ (solid line): estimated aortic pressure. V: control input voltage. r: reference value.

9.9 mmHg finally. This abnormal high $LVPED$ in 8 heart cycles is not expected to cause any damage to the human circulatory system because the upper limit 15 mmHg for $LVPED$ is a long-term limit. The numerical values for P_A and TPF both stabilized at nearly the normal values for a healthy person 17 heart cycles after the switch, 5.4 L/min and 95.2 mmHg, respectively. The mean estimation error for aortic pressure was about 5.3 mmHg in suction, and 0.4 mmHg after suction reverse. The relatively large estimation error in suction is due to the deviation of mock circulatory loop from the state-space model (5.6) in suction situation. In the process of suction reverse, this deviation decreases, as is the estimation error. In brief, the designed LVAD physiological control system has proven its ability to reverse the suction of the left ventricle.

IX. Conclusion

The design of a physiological controller for a permanent LVAD is described in this paper. With the single-equation model (5.6) derived for the human circulatory system, the adaptive estimation and control methods have been applied in the controller design.

The computer simulation and mock circulatory loop test consistently show good controller performance in the variation of

physiological states or activities, in terms of aortic pressure estimation error (less than 2 mmHg), restoring abnormal hemodynamic variables back to normal range, etc. As shown in Figs. 5 and 8, TPF was maintained at 5–6 L/min in rest, and could be increased properly in exercise; $LVPED$ was maintained between -3 mmHg and 15 mmHg. The designed physiological controller also proved its ability to reverse the collapse of the left ventricle in the mock loop test.

In summary, the critical objectives of physiological controller for permanent LVAD were successfully achieved. The significance of this work lies in several of the following aspects: 1) the state–space model (5.6) eliminated the large disturbance of left ventricle; 2) the adaptive estimation algorithm tracked the time-varying parameter TPR accurately; 3) the adaptive optimal controller adjusted controller gain corresponding to the estimate TPR, thus regulating the mean aortic pressure properly; and 4) finally, the update of the reference value $r(t)$ through the f function was able to represent the variation of the body's needs.

A series of animal tests, which replicate the human circulatory system more accurately than the Matlab model and the mock circulatory loop, is the next step to verify the designed LVAD physiological controller performance. Animal tests can also provide important data to optimize the function f in (6.7) to better represent the physiological needs. A better signal processing method to obtain y' from $\Delta P|_{t=t_s}$ may minimize the delay effect in aortic pressure estimation, thus improving the performance of the physiological controller. Although this controller is designed for LEV VAD under development, it can also be applied on any LVAD with a pump head and motor signals available.

APPENDIX
PARAMETERS AND THEIR VALUES

FULL-ORDER SYSTEM MODEL OF [4, FIG. 1]

$C_{\rm LV}$: capacitance of left ventricle; $C_{\rm RV}$: capacitance of right ventricle; $C_2 = 0.83$ ml/mmHg: capacitance of aorta; $C_3 = 1$ ml/mmHg: capacitance of systemic artery; $C_o = 40$ ml/mmHg: capacitance of vein for the nonmuscular organs; $C_m = 10$ ml/mmHg: capacitance of muscle vein; $C_4 = 20$ ml/mmHg: capacitance of central vein; $C_6 = 2.5$ ml/mmHg: capacitance of pulmonary artery; $C_7 = 10$ ml/mmHg: capacitance of pulmonary vein/left atrium; $R_{AV} = 0.004$ mmHg/(ml/s): resistance of aortic valve; $R_9 = 0.12$ mmHg/(ml/s): resistance of peripheral vessels; $R_{m_1} = 8.1$ mmHg/(ml/s): skeletal muscle vascular resistance; $R_{m_2} = 0.2$ mmHg/(ml/s): skeletal muscle venous resistance; $R_{o_1} = 0.95$ mmHg/(ml/s): vascular resistance of nonmuscular organs; $R_{o_2} = 0.023$ mmHg/(ml/s): venous resistance of nonmuscular organs; $R_{11} = 0.004$ mmHg/(ml/s): resistance of systemic vein and right atrium; $R_{12} = 0.002$ mmHg/(ml/s): resistance of pulmonary great artery; $L_9 = 0.01$ mmHg/(ml/s^2): inductance of blood in peripheral circulation; $L_{13} = 0.002$ mmHg/(ml/s^2): inductance of blood in pulmonary circulation; $R_{13} = 0.1$ mmHg/(ml/s): pulmonary vascular resistance; $R_{14} = 0.004$ mmHg/(ml/s): resistance of pulmonary vein and left atrium; D_1: aortic valve; D_{11}: tricuspid valve; D_{12}: pulmonary valve; D_2: Mitral valve; D_m: venous valve on skeletal muscle; D_{o_1}, D_{o_2}: venous valve on nonmuscular organs; TPF: total peripheral flow; Q_{m_1}: muscle vascular flow rate; Q_{m_2}: muscle venous flow rate.

REDUCED-ORDER SYSTEM MODEL OF [4, FIG. 2]

$C^d_{\rm LV} = 10$ ml/mmHg: left ventricular diastolic capacitance; $C_A = 1.83$ ml/mmHg: arterial capacitance; $C_V = 70$ ml/mmHg: venous capacitance; $L = 0.01$ mmHg/(ml/s): inductance of peripheral circulation; $R_{AV} = 0.002$ mmHg/(ml/s): resistance of aortic valve; $R_V = 0.004$ mmHg/(ml/s): resistance of vein; $\bar{\delta}_m < 5$ mmHg: the mean pressure disturbance generated by muscle pump function in exercise; $\delta_P \approx 5$ mmHg: pressure disturbance generated by pulmonary circulation. The compliance of the left ventricle during systole follows the relation described in (2.1).

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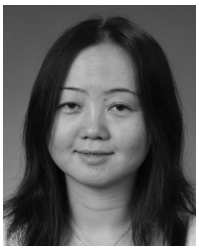
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